

Hao Luo

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🌐 <https://lighttet.github.io>

🎓 <https://scholar.google.com/citations?user=TwuNaTYAAAAJ>

EDUCATION

Peking University, School of Computer Science 2022 - 2027 expected

Ph.D. Student in Computer Applied Technology; GPA: 3.8/4.0;

Research Focus: Embodied AI, Multimodal Foundation Models; Advisor: Zongqing Lu

Peking University, School of Electronics Engineering and Computer Science 2018 - 2022

B.S. in Computer Science; GPA: 3.6/4.0

EXPERIENCES

ByteDance 2026.6 - present

Embodied AI Algorithm Intern, Seed - Multimodal Interaction and World Model Team

Beijing BeingBeyond Technology Co., Ltd. 2025.4 - 2026.6

Research Intern, Being Foundation Model Team

- ♦ Developing the Being-H series of [embodied foundation models](#), progressively scaling embodied pretraining from thousands to tens of thousands of hours of human videos.
- ♦ Studying [scalable VLA](#) and [latent world-action models](#), both leading to papers.

Beijing Academy of Artificial Intelligence (BAAI) 2023.1 - 2025.3

Research Intern, Multimodal Interaction Group

- ♦ Studied [video pretraining](#) for policy learning and video understanding [VLM](#), both leading to papers.

RECENT WORK

Being-H0.5: Scaling Human-Centric Robot Learning for Cross-Embodiment Generalization

[\[Co-first author, first-listed\]](#) Extends Being-H0's human-video pretraining to **35,000+** hours of videos and builds a unified pretraining framework across human data and 30 robot embodiments, enabling downstream cross-embodiment policies.

Rethinking Visual-Language-Action Model Scaling: Alignment, Mixture, and Regularization

[\[Co-first author\]](#) Systematically studies action alignment, heterogeneous data mixing, and regularization in VLA scaling, clarifying how these factors affect scalability and robustness.

Being-H0.7: A Latent World-Action Model from Egocentric Videos

[\[Co-first author, first-listed\]](#) Builds an implicit world model through joint alignment in latent space, enabling a imagine-then-act pattern without generating future video frames. Large-scale heterogeneous pretraining improves performance in dynamic scenes and long-horizon tasks.

Embodied AI / VLA**Being-H0: Vision-Language-Action Pretraining from Large-Scale Human Videos**

[ICML 2026, Co-first author, first-listed] The earliest work to use large-scale human videos for VLA pre-training, treating human hands as a universal manipulator and serializing human motions through structured hand-motion tokenization, transferring human priors to downstream robotic dexterous manipulation.

Joint-Aligned Latent Action: Towards Scalable VLA Pretraining in the Wild

[CVPR 2026, First author] Developing JEPT, JALA scales VLA pretraining by aligning hidden states with inverse dynamics to learn unified latent actions from both labeled and unannotated videos, reducing data bottlenecks.

Unmasking the Illusion of Embodied Reasoning in Vision-Language-Action Models

[ECCV 2026] Haiweng Xu, Sipeng Zheng, **Hao Luo**, Wanpeng Zhang, Ziheng Xi, Zongqing Lu†

Transport Discrepancy as a Reliability Signal for Vision-Language-Action Models

[ECCV 2026] Wanpeng Zhang, Ye Wang, **Hao Luo**, Haoqi Yuan, Yicheng Feng, Sipeng Zheng, Qin Jin, Zongqing Lu†

Spatial-Aware VLA Pretraining through Visual-Physical Alignment from Human Videos

[CVPR 2026] Yicheng Feng, Wanpeng Zhang, Ye Wang, **Hao Luo**, Haoqi Yuan, Sipeng Zheng, Zongqing Lu†

OpenT2M: No-frill Motion Generation with Open-source, Large-scale, High-quality Data

[CVPR 2026] Bin Cao, Sipeng Zheng, **Hao Luo**, Boyuan Li, Jing Liu, Zongqing Lu†

Video Representation for Policy Learning**Learning Video-Conditioned Policy on Unlabelled Data with Joint Embedding Predictive Transformer**

[ICLR 2025, First author] Inspired by JEPA, JEPT learns predictive visual representations of task-relevant transitions through joint embedding prediction, enabling the earliest JEPA-style learning in policy learning.

Pre-trained Visual Dynamics Representations for Efficient Policy Learning

[ECCV 2024, First author] PVDR learns abstract visual dynamics priors from unlabeled videos and adapts them for efficient policy learning in domain-gapped tasks.

Reinforcement Learning Friendly Vision-Language Model for Minecraft

[ECCV 2024] Haobin Jiang*, Junpeng Yue*, **Hao Luo**, Ziluo Ding, Zongqing Lu†

Vision-Language Models**OpenMMEgo: Enhancing Egocentric Understanding for LMMs with Open Weights and Data**

[NeurIPS 2025, First author] Builds an 8.2M-sample egocentric video QA dataset and introduces semantic-aware visual token compression to improve VLMs in egocentric video understanding.

Unified Multimodal Understanding via Byte-Pair Visual Encoding

[ICCV 2025 Highlight] Wanpeng Zhang, Yicheng Feng, **Hao Luo**, Yijiang Li, Zihao Yue, Sipeng Zheng, Zongqing Lu†

VideoOrion: Tokenizing Object Dynamics in Videos

[ICCV 2025] Yicheng Feng*, Yijiang Li*, Wanpeng Zhang, Sipeng Zheng, **Hao Luo**, Zihao Yue, Zongqing Lu†

Reinforcement Learning**A Survey on Transformers in Reinforcement Learning**

[TMLR 2023] Wenzhe Li*, **Hao Luo***, Zichuan Lin*, Chongjie Zhang†, Zongqing Lu†, Deheng Ye†

HONORS & SERVICES

- ◆ Lingjun Pilot Scholarship, Peking University (PKU) 2025
- ◆ Merit Student, by Peking University (PKU) 2024-2025
- ◆ Academic Excellence Award, by Peking University (PKU) 2019-2021
- ◆ Social Work Award, by Peking University (PKU) 2018-2019
- ◆ Reviewer: ICML, NeurIPS, ICLR, CVPR, ICCV, ECCV, BMVC
- ◆ Teaching Assistant: Design and Analysis of Algorithms, PKU (Spring 2021, Spring 2022); Deep Reinforcement Learning, PKU (Spring 2024)